Part C - Deriving Attention Gradients by hand -

Setup: $S = \dfrac{Q \cdot K^T}{\sqrt{d_k}}$, $P = \text{softmax}(S)$, $A = P \cdot V$

17) $\dfrac{\partial A}{\partial V} = ?$

$A = P.V \Rightarrow A_{ij} = \sum P_{ik} V_{kj}$ (element-wise)

$$\frac{\partial (A_{ij})}{\partial V_{mn}} = P_{im} \frac{\partial \sum V_{kj}}{\partial V_{mn}}$$

$$= P_{im} \frac{\partial V_{mj}}{\partial V_{mn}} \quad (k=m \text{ term survives}).$$

$$\frac{\partial V_{mj}}{\partial V_{mn}} = \delta_{jn} \Rightarrow \boxed{\frac{\partial (A_{ij})}{\partial V_{mn}} = P_{im} \delta_{jn}}$$

In matrix-form, $\boxed{\dfrac{\partial A}{\partial V} = P}$ → gradients flowing into V are weighted by P.

18) Softmax Jacobian - $p = \text{softmax}(s)$

$$P_i = \frac{e^{S_i}}{\sum_k e^{S_k}}$$

Jacobian: $\dfrac{\partial P_i}{\partial S_j} = ?$

Case-1 :- $i = j \Rightarrow \dfrac{\partial P_i}{\partial S_i} = \dfrac{\partial}{\partial S_i}\left[\dfrac{e^{S_i}}{\sum_k e^{S_k}}\right]$

$$= \frac{\sum_k e^{S_k}\left[e^{S_i}\right] - e^{S_i}\left[e^{S_i}\right]}{\left(\sum_k e^{S_k}\right)^2}$$

$$= p_i(1 - p_i)$$

Case-2:- $i \neq j \Rightarrow \dfrac{\partial p_i}{\partial s_j} = \dfrac{\partial}{\partial s_j}\left[\dfrac{e^{s_i}}{\sum_k e^{s_k}}\right]$

$$= \frac{\sum_k e^{s_k}\left[e^{s_i} \underbrace{\cancel{\frac{\partial s_i}{\partial s_j}}}_{0}\right] - e^{s_i}\left[e^{s_j} \overbrace{\cancel{\frac{\partial s_j}{\partial s_j}}}^{1}\right]}{\left(\sum_k e^{s_k}\right)^2}$$

$$= -p_i p_j$$

$$\frac{\partial p_i}{\partial s_j} = \begin{cases} p_i(1-p_i) & i=j \\ -p_i p_j & i \neq j \end{cases}$$

$$\delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\Rightarrow \boxed{\frac{\partial p_i}{\partial s_j} = p_{ij}(\delta_{ij} - p_{ij}) \quad \forall \; i,j}$$

19) $\dfrac{\partial A}{\partial Q} = ?$ $\qquad \dfrac{\partial A}{\partial Q} = \dfrac{\partial A}{\partial P} \cdot \dfrac{\partial P}{\partial S} \cdot \dfrac{\partial S}{\partial Q}$

→ $S = \dfrac{Q \cdot K^T}{\sqrt{d_k}} \Rightarrow \dfrac{\partial S}{\partial Q} = \dfrac{K^T}{\sqrt{d_k}}$

→ $\dfrac{\partial P}{\partial S} = P(\delta - P)$ (matrix form)

→ $A = PV \Rightarrow \dfrac{\partial A}{\partial P} = V$; $\dfrac{\partial A_{ab}}{\partial P_{ai}} = V_{ib}$

$$\Rightarrow \frac{\partial P_{ai}}{\partial S_{aj}} = P_{ij}(\delta_{ij} - P_{ij})$$

$$\Rightarrow S_{aj} = \frac{\sum Q_{ax} K_{jx}}{\sqrt{d_k}}$$

$$\frac{\partial S_{aj}}{\partial Q_{mn}} = \frac{1}{\sqrt{d_k}} \cdot \sum_x K_{jx} \cdot \frac{\partial Q_{ax}}{\partial Q_{mn}}$$

$$= \frac{1}{\sqrt{d_K}} \cdot \sum_x K_{jx} \cdot \delta_{am} \cdot \delta_{xn} \cdot$$

$$= \frac{1}{\sqrt{d_K}} \cdot \delta_{am} \cdot K_{jn} \quad (x = n \text{ term only lives})$$

$$\Rightarrow \frac{\partial S_{aj}}{\partial Q_{mn}} = \frac{1}{\sqrt{d_K}} \cdot \delta_{am} K_{jn}$$

$$\Rightarrow \frac{\partial A_{ai}}{\partial Q_{mn}} = \sum_i V_{ib} \sum_j P_{ij}(\delta_{ij} - P_{ij}) \cdot \frac{1}{\sqrt{d_K}} \cdot \delta_{am} K_{jn}$$

$$\boxed{\frac{\partial A_{ai}}{\partial Q_{mn}} = \frac{\delta_{am}}{\sqrt{d_K}} \sum_i \sum_j V_{ib} P_{ij} (\delta_{ij} - P_{ij}) \cdot K_{jn}}$$

20) Intrepretation -

When the logits have very high magnitudes ($S = Q \cdot K^T$ is a huge number), the softmax outputs are very peaked. One probability becomes very close to 1 while the others remain nearly zero. Hence,

$$\frac{\partial P_i}{\partial S_j} = P_{ij}(\delta_{ij} - P_{ij}) \begin{cases} \rightarrow P_i(1 - P_i) \\ \rightarrow -P_i P_j \end{cases}$$

$\frac{\partial l_i}{\partial S_j}$ gradients become very small so, the gradient flowing back through softmax vanishes. This is a problem as the model cannot learn properly. This is corrected by introducing a $\frac{1}{\sqrt{d_K}}$ factor. As S_j (dot product) rises with dimension d_K, this scaling keeps the softmax stable and the gradients meaningful.